

DHEERAJ KUMAR

AI Software Engineer | Full Stack Developer | Generative AI

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in **Artificial Intelligence, Computer Vision, Generative AI, and Full Stack Development**. Experienced in developing **AI-powered applications** using **React.js, Node.js, Express.js, MongoDB, PyTorch, LangChain, OpenAI API, Gemini API, and Hugging Face**. Strong foundation in **Machine Learning**, backend development, **REST APIs**, authentication, deployment, and scalable software engineering with a passion for building production-ready **AI solutions**.

SKILLS

Databases :	MongoDB, MySQL
AI/ML & Generative AI :	Machine Learning, Deep Learning, Computer Vision, Generative AI, LangChain, PyTorch, YOLOv11, Hugging Face, OpenAI API, Gemini API
Languages :	Hindi, English
Programming Languages :	Python, JavaScript, Java, SQL
Soft Skills :	Problem Solving, Decision Making, Teamwork, Critical Thinking, Adaptability, Time Management
Tools & Platforms :	Git, GitHub, Docker, Linux, AWS, Postman, Render, Vercel, VS code
Web Development :	React.js, Node.js, Express.js, MongoDB, HTML5, CSS3, Tailwind CSS, REST APIs, JWT Authentication
Core CS :	Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming(OOP)

PROJECTS

AI Real Estate Agent | 🐙 Github ⚙️ Vercel

Developer

- **Developed** an **AI-powered real estate platform** enabling natural-language property search and conversational assistance.
 - **Integrated OpenAI and Gemini APIs** to provide personalized property recommendations based on budget, location, and user preferences.
 - **Built** a secure **MERN architecture**, role-based access control, and responsive **React frontend**.
 - **Designed and implemented REST APIs** for authentication, properties, favorites, and AI chat functionality.
 - **Deployed** the application using **Vercel, Render, and MongoDB Atlas** with **Git/GitHub version control**.
- Technologies / Tools Used : Node.js, MongoDB, OpenAI API, Gemini API, Tailwind CSS, React, Express*

Brain Tumor Segmentation using Deep Learning | 🐙 Github

Developer

- **Developed** a **PyTorch-based brain tumor segmentation model** achieving approximately **92% Dice Score**.
 - **Built** a **Streamlit application** for real-time MRI upload, visualization, and prediction.
 - **Integrated LangChain** with **LLM APIs** to generate **AI-assisted medical reports**.
 - **Automated** report generation and email delivery using **SMTP**, reducing manual effort by over **90%**.
- Technologies / Tools Used : Python, PyTorch, Streamlit, LangChain, OpenAI API, Hugging Face*

PPE Violation Detection System | 🐙 Github

Developer

- **Fine-tuned YOLOv11** on 8,000+ annotated images achieving approximately **91% mAP@0.5**.
 - **Built** a real-time dashboard capable of processing live video streams at **25–30 FPS**.
 - **Implemented** automated detection of helmets, gloves, and safety equipment for workplace monitoring.
- Technologies / Tools Used : Python, YOLOv11, OpenCV, Streamlit*

Airbnb Clone | 🌐 Website 🐙 Github

Developer

- **Developed** a **full-stack accommodation booking platform** inspired by **Airbnb**.
 - **Implemented** authentication, property listings, booking workflow, reviews, and host dashboard.
 - **Designed and developed 10+ REST APIs** following **MVC architecture** and optimized **MongoDB queries**.
 - **Deployed** the application using **Render, Vercel, Git, and GitHub**.
- Technologies / Tools Used : React.js, Node.js, Express.js, MongoDB, JWT*

EDUCATIONS

Dr. A.P.J. Abdul Kalam Technical University | Uttar Pradesh

Aug 2023 - May 2027

B.Tech in Computer Science Engineering

CERTIFICATIONS

- Deloitte Technology Job Simulation (Forage)
- Generative AI Essentials – TCS iON
- AI & Cybersecurity Awareness – TCS iON
- Linux & AWS Cloud Computing (36 Hours)
- CodeChef Data Structures
- Network Fundamentals (Infosys Springboard)